



Autonomous Boat Craft for Emergency Naval Delivery

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Introduction & Problem Statement

Coastal disasters can destroy ports, roads, and docks, leaving emergency supplies stranded offshore while affected populations remain unreachable.

Crewed rescue boats expose operators to hazardous conditions, drones are limited by payload and weather, and most autonomous surface vessels are not designed for near-shore emergency delivery.

To address this gap, we developed a solar-powered autonomous monohull surface vessel for repeated delivery of small emergency payloads in shallow coastal environments.



System Overview



Engineering Analysis

Propulsion

$TWR = \frac{T}{W}$
Thrust is sized to exceed drag and ensure forward motion.

T: total thrust from motors (N)
W: total weight of the vessel (including payload) (N)

Payload & Displacement

Buoyancy must balance total weight for stable flotation.

$$W = pgV$$

W: total weight of vessel (N)
P: density of water (kg/m³)
V: displaced water (m³)
g: gravity(m/s²)

Drag and Resistance

Hydrodynamic drag determines propulsion requirements.

$$F_D = \frac{1}{2} \rho C_D A v^2$$

Steering and Maneuvering

Differential thrust generates torque for turning

$$\tau = (T_R - T_L)d \quad \tau = I\alpha$$

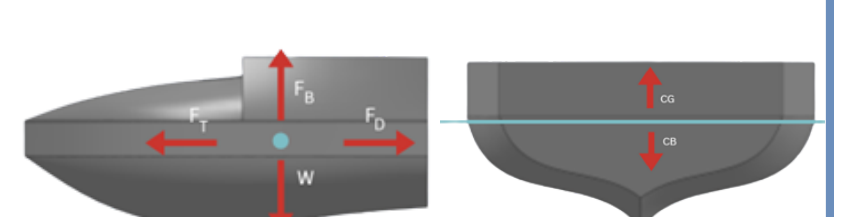
Power Budget

System power is limited by battery and supplemented by solar input

$$I_{total} = \frac{P_{motors} + P_{logic} - P_{solar}}{V_{system}}$$

Current (<30A)
V: 14.8V
P: 300W (motors)

FBDs



Calculated Values

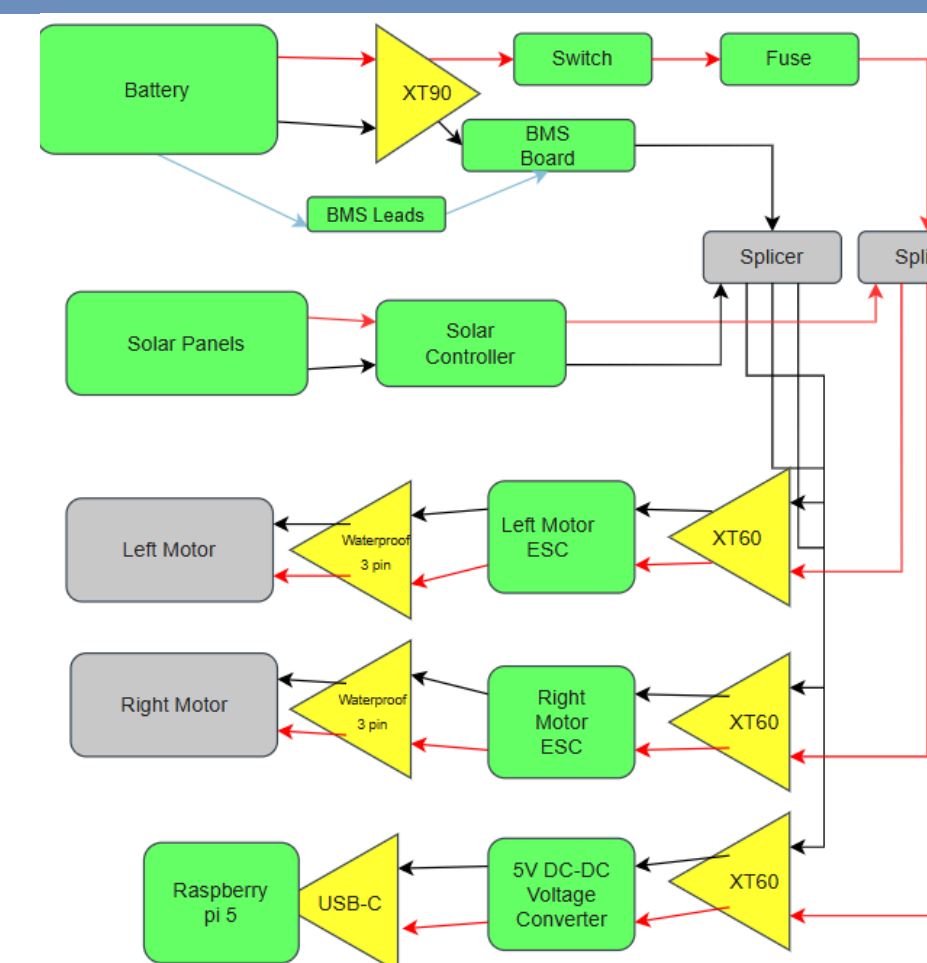
Max continuous system power: 444 W
Solar contribution: 3.4% battery capacity
Current displacement utilization: 56.1%
Max buoyant support: 34.4 lb
Battery energy: 296 Wh
Reserve buoyancy: 15.1 lb

Project Objectives

Objective	Target/Intent
Proof of Concept	Demonstrate a fully integrated autonomous delivery platform
Hull Platform	Design and fabricate a stable monohull surface vessel
Solar-Assisted Power	Supplement onboard power and extend mission endurance
Autonomous Navigation	Execute waypoint-based missions without constant operator input
Sensor Integration	Incorporate GPS, camera, Wi-Fi, and water sensing for navigation and safety
Waterproofing	Protect onboard electronics from water ingress during operation
Payload Capability	Carry a small emergency supply payload
Field Validation	Verify flotation, propulsion, control, and subsystem performance in water

Electrical Architecture

- Primary power:** 14.8V LiPo battery, managed by 30A BMS.
- Safety:** A manual disconnect switch and 40A MAXI fuse feeding into central splicers.
- Solar:** Two 12V solar panels feeding into a MPT-7210A boost controller.
- Propulsion:** Two 300W motors driven by independent 20A ESCs.
- Computing:** A 5V DC-DC buck converter to stabilize power to the Raspberry Pi 5.
- External Sensors:** Powered through 5V and 3.3V pins from Raspberry Pi 5.



Testing & Results

Test	What was evaluated	Result
Float & Stability	Buoyancy, trim, and payload-carrying stability	Maintained positive flotation and acceptable trim under payload; minor pitch/list was observed but did not prevent operation
Leak Test	Hull waterproofing under submersion	30s submersion produced zero internal water ingress
Propulsion Test	Forward motion under payload in pond conditions	Twin-thruster propulsion produced stable forward motion under payload
Steering Test	Differential-thrust maneuvering	Controlled turning was successfully demonstrated through differential thrust
Waypoint Navigation	Outdoor GPS-based autonomous navigation	Completed a 100m waypoint mission and arrived within approximately 3m of the target
Sensor Integration	GPS, camera, Wi-Fi, and water sensing	All core sensors and communication modules were successfully demonstrated
Solar Integration	On-system solar functionality	Solar subsystem operated successfully; quantitative endurance gains remain future work

Mechanical Design

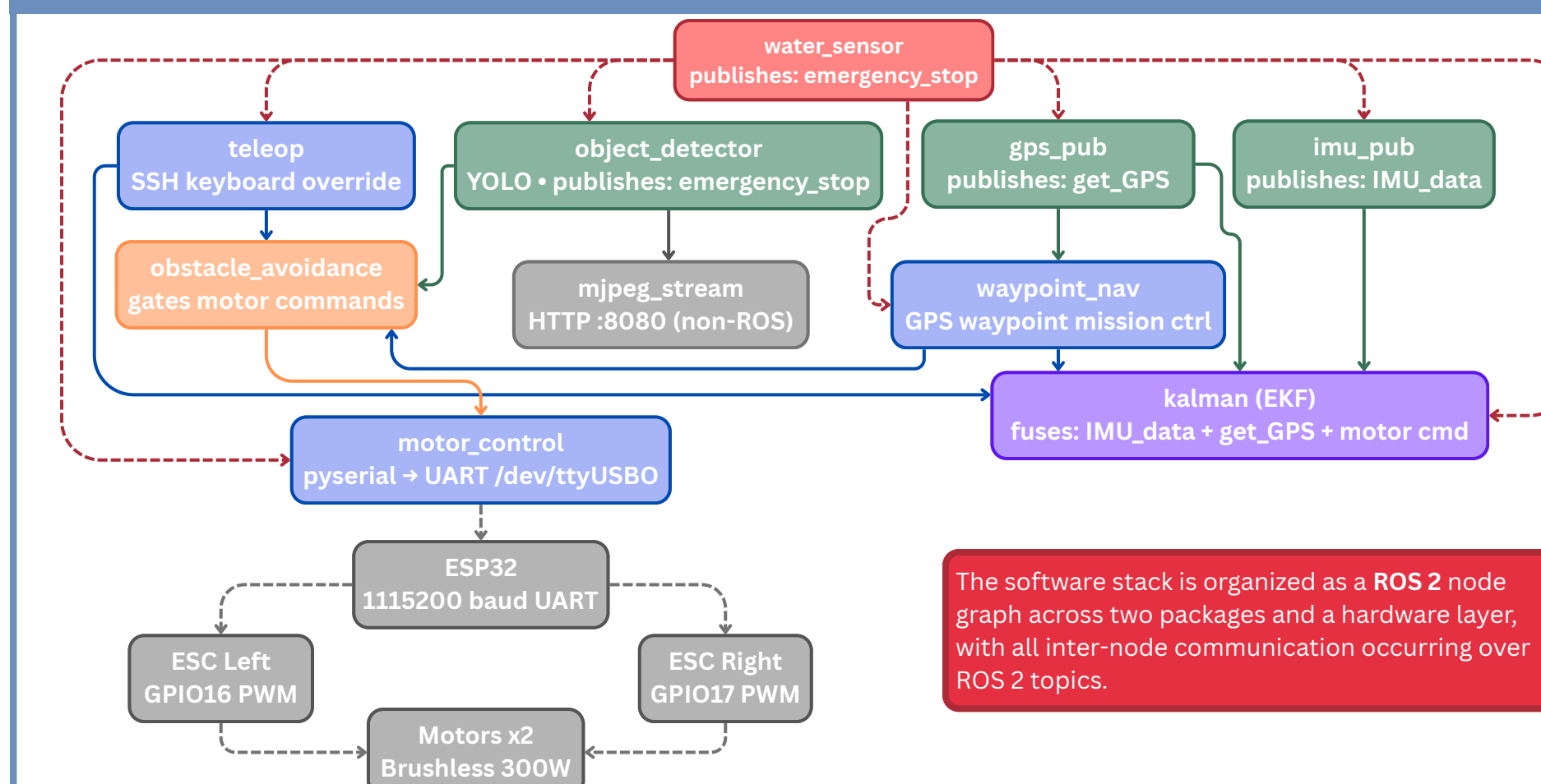
Polyurethane, Fiberglass Composite, Marine Grade Epoxy, Sanding
Primer & Paint
Marine Rubber Sealant Tape, Rubber Latches
S - Curve Semi-Displacement Monohull



Design Specifications

Dimensions: 24"L 12.7"W 9"H
Displaced Volume: 952.68 in³
Total Weight: 19.3 lb
Payload Capacity (~25% Buoyancy): 6.0 lb

Software & Autonomy



The software stack is organized as a **ROS 2** node graph across two packages and a hardware layer, with all inter-node communication occurring over ROS 2 topics.

Conclusion & Future Work

Conclusion

- Structural Integrity:** S-curve hull and custom seal provided 100% waterproof protection.
- Precision Maneuverability:** The 3-DOF enabled zero-turn navigation through debris.
- System Robustness:** Resolved power draw limits during high-torque maneuvers by requiring higher amperage limit Battery Management System

Future Work

- Intelligence:** Integrate AI/ML for advanced obstacle detection and complex 6-DOF hydrodynamic modeling.
- Power Optimization:** Implement power smoothing logic to manage current spikes and maximize solar efficiency.
- Alternate Applications:** Scale up design to be applied to localized sonar operations in maritime defense applications.